Elaina Mann

elainaymann@gmail.com | (973) 980-9936 | elainamann.me

EDUCATION

University of Michigan — **Ann Arbor, MI**
Bachelor of Science in Engineering in Computer Engineering — *May 2026*
Bachelor of Science in Engineering in Robotics — *May 2026*
Graduated *cum laude*

TECHNICAL SKILLS

Programming	C++, Python, ROS2, Verilog, Julia, MATLAB
Platforms & Tools	Altium, KiCad, Arduino, Raspberry Pi, STM, SolidWorks, Linux, Git, Lab Instrumentation
Core Competencies	Embedded Systems, PCB Design, Software Design, Machine Learning, Rapid Prototyping

WORK EXPERIENCE

Panasonic — **Newark, NJ**
Automation Engineering Intern — *Jan 2024 - Oct 2024*

- Designed an internal web application based on key stakeholder meetings to fully understand organizational requirements.
- Built application that successfully centralized document requests and allowed real-time chat communication between 200+ members of North America Panasonic departments.
- Automated SharePoint folder actions, improving audit workflow efficiency for the department.

University of Michigan Department of Performing Arts Technology — **Ann Arbor, MI**
Mechatronics Team Research Assistant — *Jan 2024 - May 2024*

- Designed C++ algorithms mapping musician movement to robot movement, resulting in intuitive robot response.
- Restructured robot electrical system to optimize power usage and prioritize safety.
- Modeled robot using taskspace and jointspace representations, creating a more streamlined mapping of musician movement to robot movement.

PROJECT EXPERIENCE

University of Michigan Robotic Submarine Team — **Ann Arbor, MI**
Software/Electrical Team Member — *Aug 2022 – May 2026*

- Lead research and implementation of the hardware, electrical, and Python-based software for a hydrophone localization system, leading to a top 5 placement at the international Robosub competition.
- Designed PCBs and CADed interior structures to improve electrical system organization and efficiency.
- Developed and wrote a Python-based navigation state machine for the autonomous submarine.

President — *May 2024 - May 2025*

- Led 30+ member robotic submarine team, aligning electrical, mechanical, and admin subteams.
- Directed weekly meetings, creating agendas and presentations to track progress and resolve blockers.
- Secured $20k+ in sponsorships, managed budgeting, and coordinated travel logistics for 15 members.

Autonomous Fruit Skewering Robot — **Ann Arbor, MI**
Autonomous Robotics Term Project — *Jan 2026 – May 2026*

- Designed and 3D printed dual-arm robot components and 360° rotating base to support bimanual workspace coverage for food skewering tasks
- Implemented autonomous movement behaviors using vision-based imitation learning from leader-follower teleoperation data
- Integrated RGB-D camera perception pipeline to detect food items and skewer holder positions, feeding observations into a closed-loop visuo-motor control system

Wireless Blindspot Detection System for Semi-Trucks — **Ann Arbor, MI**
Advanced Embedded System Term Project | Sponsored by Infineon — *Aug 2025 - Dec 2025*

- Designed and soldered two custom PCBs featuring ESP32 microcontrollers, ultrasonic sensors, boost converters, and LiPo battery management circuits.
- Implemented Bluetooth communication architecture enabling Barnacle sensor units to report vehicle detection status.
- Developed iOS companion app supporting device configuration, driving mode notifications, and live diagnostic logging.